

ICAD RT-4 • Set 8 — Answer Sheet

Exam 12-10-2026 • 75 Q • 300 marks • 3 hours • +4 correct, -1 wrong, 0 blank

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Physics

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Numerical — write the value

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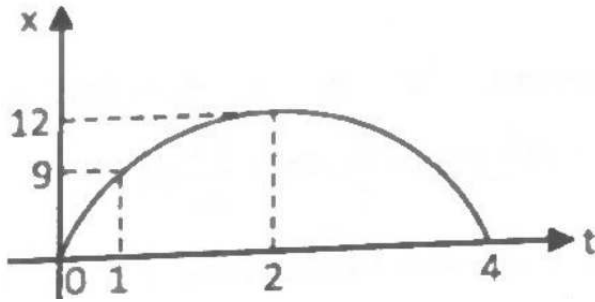
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Bubbles — single correct • Box — write the value • each subject out of 100

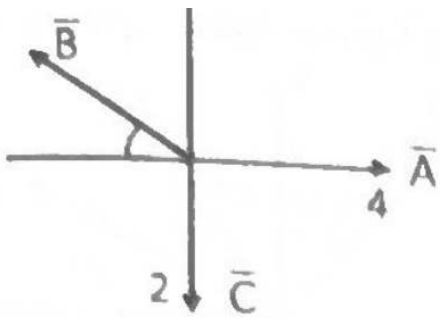
Physics

PART A • Q1–20 • SINGLE CORRECT

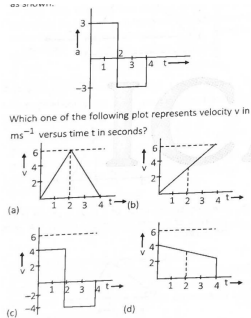
1. P2L3V6Q1 Curve $x = 12t - 3t^2$. Find slope at $t = 3$. (a) 9 (b) 6 (c) -9 (d) -6



2. P3L3L4V5Q5 $\vec{A} + \vec{B} + \vec{C} = 1\hat{j}$ with $\vec{A} = 4$ (x-axis), $\vec{C} = 2$ (down), $\tan\theta = 3/4$: write $\vec{B}$ in component form; magnitude & direction. (a) $-4\hat{i} + 3\hat{j}$; 5.0, 37 deg (b) $4\hat{i} - 3\hat{j}$; 3.5, 53 deg (c) $-4\hat{i} - 3\hat{j}$; 4.5, 65 deg (d) $4\hat{i} + 3\hat{j}$; 5.0, 53 deg

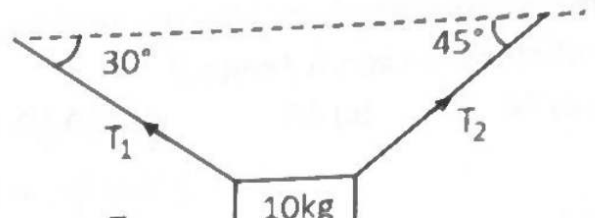


3. P3L6L7V2Q3 If vector $\vec{A} = 3\hat{i} + 4\hat{j}$ and vector $\vec{B} = \hat{i} + 2\hat{j}$, what is $\vec{A} \cdot \vec{B}$? (a) 11 (b) 10 (c) 12 (d) 14
4. P4L6V4Q1 A particle starts from rest at $t=0$ and undergoes acceleration a (ms^{-2}) vs t as shown (step graph, +3 then -3). Which plot represents velocity v (ms^{-1}) vs time t ? (a) triangle: v rises 0 to 6 at $t = 2$ s then falls to 0 at $t = 4$ s (b) straight line: v rises 0 to just over 6 at $t = 4$ s (c) step: $v = 4$ for 0-2 s then $v = -4$ for 2-4 s (d) v falls gently from 4 at $t = 0$ to about 2.5 at $t = 4$ s then drops to 0

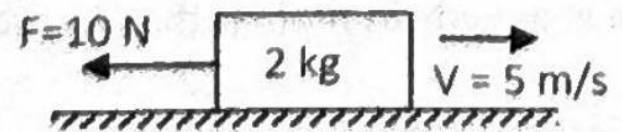


5. P4L7V4Q2 A particle is projected with velocity v so that its range on a horizontal plane is twice the greatest height attained. Range is: (a) $4v^2/5g$ (b) $4g/5v^2$ (c) $4v^3/5g^2$ (d) $4v/5g^2$
6. P4L10V5Q5 Cars A and B move towards each other; separated by 30 m, A turns left at 30° and B turns right at 60° , both measured from the line joining the cars, at the same instant, and they collide after 2 s. Identify the WRONG option about their speeds / relative velocity. (a) speed of car A is 7.5 ms^{-1} (b) speed of car B is 7.5 ms^{-1} (c) speed of car A is $7.5\sqrt{3} \text{ ms}^{-1}$ (d) speed of car A with respect to car B after taking turns is 15 ms^{-1}
7. P4L11V2Q5 Man running at 5 m/sec; raindrops appear at 45° from vertical. If actually falling vertically downwards, find velocity. (a) 5 (b) $5\sqrt{3}$ (c) $5\sqrt{2}$ (d) 4

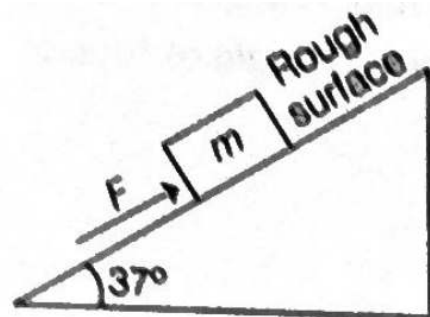
8. P5L3V4Q10 A body of mass 1 kg is suspended by two strings of lengths 3 m and 4 m from two nails on a wall, horizontal separation 5 m. Find the tensions in the longer and shorter strings. Take $g = 10 \text{ ms}^{-2}$: (a) 6N, 6N (b) 6N, 8N (c) 6N, 4N (d) 8N, 8N
9. P5L3V5Q10 A 10 kg mass hangs with strings at 30° and 45° . Find the ratio T_2/T_1 : (a) $3/2$ (b) $\sqrt{3}/2$ (c) $\sqrt{3}/2$ (d) $2/\sqrt{3}$



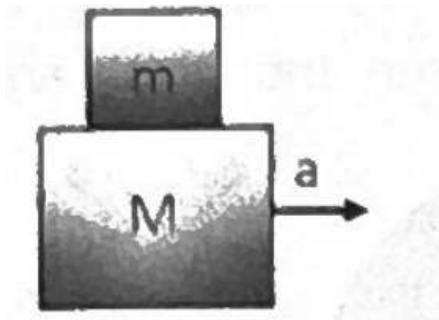
10. P6L1V5Q1 Force $\vec{F} = 6t^2 \hat{i}$; linear momentum $\vec{p}$ at $t = 0$ is 5 kg·m/s ($\hat{i}$). Find $\vec{p}$ at $t = 2$ s. (a) $16\hat{i}$ (b) $21\hat{i}$ (c) $12\hat{i}$ (d) $61\hat{i}$
11. P6L1V5Q7 Five forces on a 2 kg particle give 4 m/s^2 east; removing $\vec{F}_1$ gives 7 m/s^2 north. Acceleration if only $\vec{F}_1$ acts? (a) 16 m/s^2 (b) $\sqrt{65} \text{ m/s}^2$ (c) $\sqrt{260} \text{ m/s}^2$ (d) $\sqrt{33} \text{ m/s}^2$
12. P7L1V5SuggQ4 A 2 kg block moving rightward at $V = 5 \text{ m/s}$ is acted on by $\vec{F} = 10 \text{ N}$ directed leftward, on rough ground. Find the direction of the kinetic friction force (a) on the block, exerted by the ground (f_1), and (b) on the ground, exerted by the block (f_2).



13. P7L2V6Q7 A block of mass 10 kg is placed on a rough inclined plane of inclination 37° . A force $\vec{F}$ is applied on the block along the plane; $\mu_k = 0.2$ and $\mu_s = 0.3$ ($\sin 37^\circ = 0.6$, $g = 10 \text{ m/s}^2$). Match List-I (a: $F = 40 \text{ N}$, b: $F = 80 \text{ N}$, c: $F = 30 \text{ N}$, d: $F = 90 \text{ N}$) with List-II (p: friction is static, q: friction is kinetic, r: friction acts upward, s: friction acts downward, t: friction is zero). (a) a-pr, b-ps, c-qr, d-q (b) a-qr, b-pr, c-qr, d-q (c) a-pr, b-ps, c-pr, d-st (d) a-ps, b-st, c-qr, d-ps



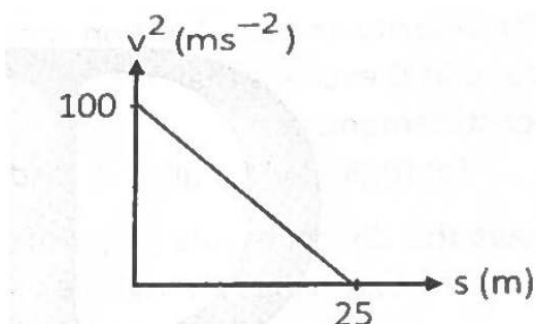
14. P7L4V3Q4 A block of mass m is placed on top of another block of mass M . The coefficient of friction between them is μ . What is the maximum acceleration with which M may move so that m also moves along with it? (a) μg (b) g/μ (c) μ^2/g (d) g/μ^2



15. P8L1V4Q6 A force $\vec{F} = (3x\hat{i} + 4\hat{j})$ newton (where x is in metres) acts on a particle which moves from a position $(2 \text{ m}, 3 \text{ m})$ to $(3 \text{ m}, 0 \text{ m})$. Then the work done is: (a) 7.5 J (b) -12 J (c) -4.5 J (d) +4.5 J
16. P8L2V5Q3 A locomotive of mass m starts moving so that its velocity varies according to the law $v = k\sqrt{s}$ where k is constant and s is the distance covered. Find the total work performed by all the forces which are acting on the locomotive during the first t seconds after the beginning of motion. (a) $W = 1/8 m k^4 t^2$ (b) $W = 1/4 m^2 k^4 t^2$ (c) $W = 1/4 m k^4 t^4$ (d) $W = 1/8 m k^4 t^4$

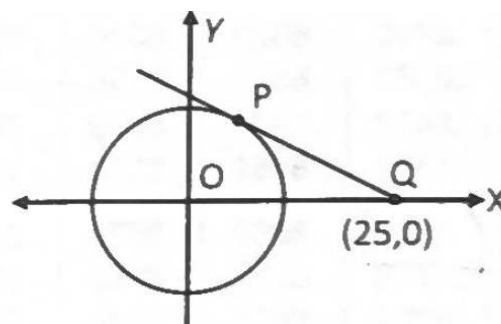
PART B · Q21–25 · NUMERICAL VALUE

21. P8L2VBoardQ1 A particle of mass 12 kg is acted upon by a force $f = (100 - 2x^2)$ where ' f ' is in newton and ' x ' is in metre. Calculate the work done by this force in moving the particle $x = 0$ to $x = +10 \text{ m}$. What will be the speed at $x = 10 \text{ m}$ if it starts from rest? — (Board)
22. P4L6V5Q10 A particle moves on a straight line whose v^2 vs displacement graph is shown ($v^2 = 100$ at $s = 0$, falling to 0 at $s = 25$). Mark the correct option(s) about when it stops, its acceleration, and the shape of its v - t and s - t graphs



17. P8L6V3Q5 A particle is released from a height S . At a certain height its kinetic energy is three times its potential energy. The height and speed of the particle at that instant are respectively (a) $S/4, 3gS/2$ (b) $S/2, \sqrt{(3gS)/2}$ (c) $S/4, \sqrt{(3gS)/4}$ (d) $S/4, \sqrt{(3gS)/2}$

18. P1L3V4Q5 Circle $r = 15 \text{ m}$ centred at origin; tangent QP from $Q(25, 0)$; length PQ? (a) 20 m (b) 25 m (c) 15 m (d) cannot be determined



19. P4L1V5Q3 A body traverses one third of the total distance with velocity 4 m/s. The remaining part is covered with velocity 2 m/s for half the time and 6 m/s for the other half. The mean velocity averaged over the whole time is: (a) 5 (b) 4 (c) 4.5 (d) 3.5 m/s
20. P8L3V3Q1 Establish relation between momentum and kinetic energy. What is the percentage change in the kinetic energy of a body if its momentum is increased by 2%? (keeping mass constant) (write the value)

23. P6L6V2Q2 2 kgf applied at one end of a horizontal spring balance and 2 kgf at the other end, opposite direction, simultaneously. Reading (in kgf)?
24. P2L2V3Q4 If $x = 1/\sqrt{(t^2 - 3)}$ then find dx/dt at $t = 2$
25. P7L1V5Q5 A 1 kg block (wood on wood: $\mu_s = 0.25, \mu_k = 0.20$) is pulled by $F(t) = 6t \text{ N}$. After motion starts, find its speed at $t = 2.0 \text{ s}$. Use $g = 9.8 \text{ m/s}^2$. Round to two decimal places; answer in m/s.

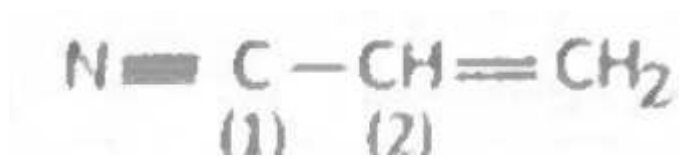
Chemistry

PART A · Q1–20 · SINGLE CORRECT

1. C2L1V6Q3 Assertion: Cathode rays consist of identical negatively charged particles irrespective of the nature of the gas or the cathode material used. Reason: In the discharge tube, any gas taken inside produces electrons that carry the same charge and mass, proving that electrons are universal constituents of all atoms. (a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.
2. C2L4V3Q3 Radius of 2nd orbit of H = $2.12 \text{ \AA}$ — radius of the 3rd orbit ($n=3$). (a) $4.77 \text{ \AA}$ (b) $9.54 \text{ \AA}$ (c) $6.36 \text{ \AA}$ (d) $3.18 \text{ \AA}$
3. C2L7V6Q9 A hydrogen atom is excited by giving it 8.4 eV of energy. The number of spectral lines emitted is. (a) none (b) two (c) three (d) four

4. C2L9V3Q1 For $n=3$ and $\ell=1$, how many orbitals are possible? (a) 1 (b) 3 (c) 5 (d) 9
5. C3L8V3Q3 Molarity of 4.9 g H_2SO_4 in 250 mL ($\text{H}_2\text{SO}_4 = 98$). (a) 0.1 M (b) 0.2 M (c) 0.3 M (d) 0.4 M
6. C3L9V3Q5 Molarity of 2 N H_2SO_4 (Basicity = 2). (a) 1 M (b) 2 M (c) 3 M (d) 4 M
7. C4L2V5Q10 According to Slater's rule, correct order of Z_{eff} on valence shell electron is: Select the correct option(s) (a) $\text{Fe} > \text{Fe}^{2+} > \text{Fe}^{3+}$ (b) $\text{N}^{3-} < \text{O}^{2-} < \text{F}$ (c) $\text{Na}^+ < \text{Mg}^{2+} < \text{Al}^{3+}$ (d) $\text{Ti}^{2+} < \text{V}^{3+} < \text{Mn}^{5+}$
8. C4L4V5Q2 Arrange S, P and As in order of increasing ionisation energy. (a) $\text{S} < \text{P} < \text{As}$ (b) $\text{P} < \text{S} < \text{As}$ (c) $\text{As} < \text{S} < \text{P}$ (d) $\text{As} < \text{P} < \text{S}$

9. C4L6V4Q3 As we move from left to right, the electronegativity increases. An atom which is highly electronegative has (a) large size (b) low electron affinity (c) high ionisation enthalpy (d) low chemical reactivity
10. C5L3V3Q1 Identify hybridisation of central atom in CH_4 and determine bond angle. (a) sp^3 , 109.5° (b) sp^2 , 120° (c) sp , 180° (d) dsp^3 , 90°
11. C5L1V5Q9 The bond between carbon atom (1) and carbon atom (2) in compound $\text{N}\equiv\text{C}(1)-\text{C}(2)\text{H}=\text{CH}_2$ involves the hybrid orbitals: (a) sp and sp^2 (b) sp^2 and sp^3 (c) sp and sp^3 (d) sp and sp



12. C5L1V5Q3 Although both HF and HCl are hydrogen halides, HF has a much higher boiling point because: (a) Fluorine has smaller atomic size than chlorine (b) HF molecules are heavier than HCl molecules (c) HF molecules are associated through hydrogen bonding (d) H—F bond length is greater than H—Cl bond length
13. C5L1V5Q5 Which of the following orders of bond length is correct for carbon-carbon bonds? (a) $\text{C}\equiv\text{C} > \text{C}=\text{C} > \text{C}-\text{C}$ (b) $\text{C}-\text{C} > \text{C}=\text{C} > \text{C}\equiv\text{C}$ (c) $\text{C}=\text{C} > \text{C}-\text{C} > \text{C}\equiv\text{C}$ (d) $\text{C}\equiv\text{C} > \text{C}-\text{C} > \text{C}=\text{C}$

PART B · Q21–25 · NUMERICAL VALUE

21. C1L11V2Q5 At what distance does London force drop to $1/64$ if distance doubles? (a) $2r$ (b) $3r$ (c) $4r$ (d) $1.5r$
22. C1L3V5Q9 4200 m^3 balloon, He at 1 atm & 27°C , air $d=1.177 \text{ kg/m}^3$ — mass of He. ($M_{\text{He}} = 4.00 \times 10^{-3} \text{ kg/mol}$, $R = 8.314 \text{ J/mol/K}$). (a) 682 kg (b) 750 kg (c) 800 kg (d) 900 kg
23. C3L8V4Q10 Bottle of conc. H_2SO_4 (density 1.80 g cm^{-3}) labelled 86% by weight; molarity?

Mathematics

PART A · Q1–20 · SINGLE CORRECT

1. M1L2V3Q3 Angle -1110° lies in which quadrant. (a) 1st quadrant (b) 2nd quadrant (c) 3rd quadrant (d) 4th quadrant
2. M2L1V4Q6 100 students: 55 passed Maths, 67 Physics; Physics only. (a) 33 (b) 22 (c) 45 (d) 10
3. M3L1V3Q5 k for which $(\cos x + \sin x)^2 + k \cdot \sin x \cos x - 1 = 0$ is an identity. (a) -1 (b) -2 (c) 0 (d) 1
4. M4L2V4Q2 Range of $f(x) = \sqrt{1-x} + \sqrt{x-1}$. (a) $\mathbb{R}$ (b) $\{0\}$ (c) $\mathbb{R} - \{1\}$ (d) $[1, \infty)$
5. M4L3V6Q9 $2\log y - \log x - \log(y-1) = 0 \rightarrow$ domain & range of $y(x)$. (a) the domain and range are both $(2, \infty)$ (b) the domain is $(0, \infty)$ and range is $(1, \infty)$ (c) the domain is $[4, \infty)$ and range is $(1, \infty)$ (d) the domain is $[4, \infty)$ and range is $(2, \infty)$
6. M5AL1V3Q4 If $\sin(\theta + \alpha) = \cos(\theta + \alpha)$, then $\tan \theta = ?$ (a) $1 - \tan \alpha$ (b) $(1 + \tan \alpha)/(1 - \tan \alpha)$ (c) $(1 + \sin \alpha)/(1 - \sin \alpha)$ (d) $(1 + \cos \alpha)/(1 - \cos \alpha)$
7. M5BL2V4Q4 If $\tan 69^\circ + \tan 66^\circ - \tan 69^\circ \tan 66^\circ = 2k$, then $k =$ (a) -1 (b) $1/2$ (c) $-1/2$ (d) none
8. M5CL3V4Q1 $1 + \cos 56^\circ + \cos 58^\circ - \cos 66^\circ =$ (a) $2\cos 28^\circ \cos 29^\circ \cos 33^\circ$ (b) $4\cos 28^\circ \cos 29^\circ \cos 33^\circ$ (c) $4\cos 28^\circ \cos 29^\circ \sin 33^\circ$ (d) $2\cos 28^\circ \cos 29^\circ \sin 33^\circ$
9. M6L4V4Q6 Domain of $f(x) = 1/(|x| + x)$ is (a) $(-\infty, 0)$ (b) $\mathbb{R} - \{0\}$ (c) $(0, \infty)$ (d) none

14. C5L1V5Q7 Amongst H_2O , H_2S , H_2Se and H_2Te , the one with highest boiling point is: (a) H_2O because of hydrogen bonding (b) H_2Te because of higher molecular weight (c) H_2S because of hydrogen bonding (d) H_2Se because of lower molecular weight
15. C2L5V4Q10 $E_n = -13.6(Z^2/n^2)$; Li^{2+} has -30.6 eV — in which orbit is the electron? (a) $n = 1$ (b) $n = 2$ (c) $n = 3$ (d) $n = 4$
16. C3L3V6Q1 A partially dried clay mineral contains 8% water. The original sample contained 12% water and 45% silica. The % of silica in the partially dried sample is nearly: (a) 50% (b) 49% (c) 55% (d) 47%
17. C4L4V2Q4 Calculate the 2nd IE if total energy for removing two electrons is 2000 kJ/mol and 1st IE is 500 kJ/mol . (a) 1500 (b) 1000 (c) 2000 (d) 2500
18. C5L8V3Q1 Calculate bond order and determine magnetism of O_2^- (17 electrons). (a) 1.5, Paramagnetic (b) 2, Diamagnetic (c) 2.5, Paramagnetic (d) 1.5, Diamagnetic
19. C1L6V6Q2 300 K, 1 atm; number density doubled & T halved — pressure via $P = (1/3)nmu^2$. (a) Pressure remains constant (b) Pressure increases $2\times$ (c) Pressure decreases $2\times$ (d) Pressure decreases $4\times$
20. C1L8V6Q3 1 mol: 22.40 L at 1 atm, 21.00 L at 2 atm — find $Z = V_{\text{real}}/V_{\text{ideal}}$ & deviation. (a) $Z = 0.9375$; gas shows negative deviation (b) $Z = 1.875$; gas shows positive deviation (c) $Z = 1.0667$; gas shows negative deviation (d) $Z = 0.8000$; gas shows positive deviation

24. C3L1V5Q9 10 mL $\text{C}_2\text{H}_5\text{OH}$ ($d = 0.7893 \text{ g/mL}$) and 20 mL water ($d = 0.9971 \text{ g/mL}$) are mixed to get a solution. If the density of resultant mixture is 0.9571 g/mL , then calculate the molarity of solution.
25. C1L1V4Q4 Liquid A (gap 1.0 mm, $v=0.10$, $F=2.0 \times 10^{-3}$) vs B (gap 2.0 mm, $v=0.20$, $F=8.0 \times 10^{-3}$) — ratio $\eta_B : \eta_A$. (a) 2 : 1 (b) 4 : 1 (c) 1 : 2 (d) 3 : 1

10. M6L5V5Q10 Find the range of the function $y = ((5+x)(2+x))/(1+x)$: (a) $(-\infty, 1] \cup [9, \infty)$ (b) $(-\infty, 1) \cup (9, \infty)$ (c) $(-\infty, 0) \cup (8, \infty)$ (d) Not
11. M7L2V3Q2 If $a = \log_2 3$ and $b = \log_3 10$, then (a) $a = b$ (b) $a > b$ (c) $a < b$ (d) none
12. M7L4V5SuggQ2 Solve for x : $2^{|x|} > 8$. (a) $x > 2$ (b) $x > 3$ (c) $x < 3$ (d) $x < 2$
13. M8AL3V5Q4 If the sum of three consecutive terms of an A.P. is 51 and the product of last and first term is 273, then the numbers are (a) 21, 17, 13 (b) 22, 17, 12 (c) 20, 17, 14 (d) 18, 17, 16
14. M8BL1V3Q5 Find 7th term of the sequence $\sqrt{2}, \sqrt{10}, 5\sqrt{2}, \dots$ is (a) $125\sqrt{10}$ (b) $25\sqrt{2}$ (c) 125 (d) $125\sqrt{2}$
15. M8BL1V5Q3 If a, b, c are in G.P then $(b-a)/(b-c) + (b+a)/(b+c) =$ (a) $b^2 - c^2$ (b) ac (c) ab (d) 0
16. M8CL1V4Q7 If a, b, c are in A.P., then $1/(\sqrt{a} + \sqrt{b}), 1/(\sqrt{a} + \sqrt{c}), 1/(\sqrt{b} + \sqrt{c})$ are in (a) A.P. (b) G.P. (c) H.P. (d) None
17. M1L2V5Q2 At 2:24 the smaller angle = exterior angle of n -gon; find n . (a) 4 (b) 5 (c) 6 (d) 8
18. M8AL2V5Q6 If $(3 + 5 + 7 + \dots \text{ to } n \text{ terms}) / (5 + 8 + 11 + \dots \text{ to } 10 \text{ terms}) = 7$, then the value of n is (a) 35 (b) 36 (c) 37 (d) 40

19.

M88L1V4Q4

The number which should be added to the numbers 2, 14, 62 so that the resulting numbers may be in GP is
(write the value)
20.

M8CL1V6Q2

If a, 4, b are in A.P. and a, $2\sqrt{3}$, b are in G.P., and if a, x, b are in H.P., then the value of x is (write the value)

PART B · Q21–25 · NUMERICAL VALUE

21.

M5AL3VSuggQ7

Find a and b such that the following inequality holds good for all θ : $a \leq 3 \cos \theta + 5 \sin(\theta - \pi/6) \leq b$.
22.

M6L1V5Q5

If $(2x+3)/4 - 3 < (x-4)/3 - 2$, then $x \in (-\infty, a)$; find |a|.
23.

M8AL1V3Q1

If the nth term of a sequence is given by $a_n = n^2 - n + 1$, find its 5th term.
24.

M8AL2V5Q2

The sum of first seven terms of an AP is 10 and that of next seven terms is 17. Then the value of 7d is where d is common difference of AP.
25.

M5CL4V5Q4

If $A + B + C = 180^\circ$ and $\sin A + \sin B + \sin C = k \cos(A/2) \cos(B/2) \cos(C/2)$, then k is _____.